Exam Linear Algebra (202001178)

University of Twente

November 2, 2020, 13:45-16:45h

This exam has 5 exercises.
Motivate all your answers by explaining how you got to them!
You may not use any electronic device.

1. Consider the following system of linear equations with $\alpha, \beta \in \mathbb{R}$.

$$\begin{cases} -3x_1 - x_2 + \alpha x_3 = -4 \\ -2x_1 + 2x_3 = -6 \\ -3x_2 - 6x_3 = 15 \\ 4x_1 + \alpha x_2 - 2x_3 = \beta \end{cases}$$

   (a) Determine for which values of α and β this system has no solutions, one solution or infinitely many solutions.

   In the remainder of this exercise, take $\alpha = 1$ and $\beta = 7$.

   (b) Determine the solution set of this system and write it in parametric vector form.

   (c) Let A be the coefficient matrix of the system. Determine a basis of Col$A$ and the dimension of this subspace.

2. Consider the linear system $A\mathbf{x} = \mathbf{b}$ with

$$A = \begin{pmatrix} 4 & 6 & 0 \\ 1 & 3 & 4 \\ 1 & 2 & 1 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} -2 \\ 5 \\ 1 \end{pmatrix}$$

   (a) Calculate the volume of the parallelepiped spanned by the columns of A.

   (b) Show that A is invertible and determine A^{-1}.

   (c) Determine the solution set of $A\mathbf{x} = \mathbf{b}$.

   (d) Consider the statement: "$\det(\alpha A) = \alpha \det(A)$ for all $\alpha \in \mathbb{R}$." Is this statement true or false?

3. Let $T : \mathbb{R}^2 \to \mathbb{R}^2$ be the following transformation. First, a vector (x_1, x_2) is reflected in the x_2-axis and then it is rotated about the origin through an angle of $\pi/2$ radians counterclockwise.

   (a) Determine the representation matrix of T.

   (b) Determine $\ker(T)$.

   (c) Determine if T is onto and/or one-to-one.

   (d) Determine all vectors that have image $(3, 2)$.

4. Consider the set

$$S = \left\{ \begin{pmatrix} 4 \\ 1 \\ 1 \\ -2 \end{pmatrix}, \begin{pmatrix} 6 \\ 3 \\ 2 \\ 5 \end{pmatrix}, \begin{pmatrix} 0 \\ 4 \\ 1 \\ 1 \end{pmatrix} \right\}.$$

(a) Show that if

$$\mathbf{x} \in \text{Span} \left\{ \begin{pmatrix} 5 \\ -4 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 6 \\ -17 \\ -3 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 4 \\ 1 \\ 1 \end{pmatrix} \right\}.$$

then $\mathbf{x} \in \text{Span}S$.

(b) Determine if the vectors in S are linearly independent.

(c) Determine $\mathbf{y}$ if $[\mathbf{y}]_S = (-1, 2, -3)^T$.

5. Let

$$A = \begin{pmatrix} 3 & 18 & 0 \\ 1 & 6 & 2 \\ 0 & 0 & 4 \end{pmatrix}, \qquad \mathbf{x} = \begin{pmatrix} -10 \\ -1 \\ 10 \end{pmatrix}, \qquad \mathbf{y} = \begin{pmatrix} 3 \\ 1 \\ 0 \end{pmatrix}.$$

(a) Determine if the vectors $\mathbf{x}$ and $\mathbf{y}$ are eigenvectors of A.

(b) Show that 4 is an eigenvalue of A.

(c) Determine the eigenvalues of A and their corresponding eigenvectors.

(d) Determine an invertible matrix P and a diagonal matrix D such that $D = P^{-1}AP$. Show that this latter equality holds.

Norm:

	1			2				3				4			5			
Exercise	a	b	c	a	b	c	d	a	b	c	d	a	b	c	a	b	c	d
Points	3	2	2	2	3	1	1	2	2	2	2	3	2	1	2	1	3	2

Total: $36 + 4 = 40$ points.